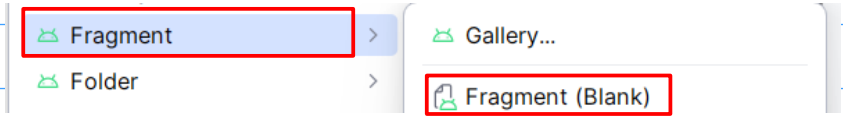
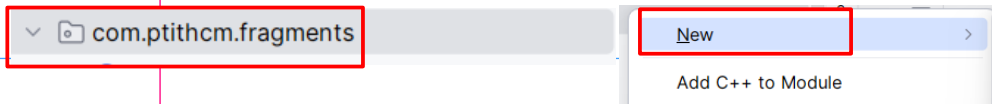


F Fragments



FragmentA

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:orientation="vertical"
    android:background="#038493"
    android:id="@+id/fragmentA"
    android:layout_height="match_parent">
    <Button
        android:id="@+id/button"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:layout_marginHorizontal="30dp"
        android:text="Click Me" />
</LinearLayout>
```

FragmentB

```
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/fragmentB"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#99ff66"
    tools:context=".FragmentB">
    <!-- TODO: Update blank fragment layout -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="30dp"
        android:textSize="30sp"
        android:id="@+id/textView"
        android:layout_gravity="center_horizontal"
        android:textAlignment="center"
        android:text="Count: 0" />
</FrameLayout>
```

```
<androidx.fragment.app.FragmentContainerView
    android:id="@+id/fragmentA"
    android:name="com.ptithcm.fragments.FragmentA"
    android:layout_weight="1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

<androidx.fragment.app.FragmentContainerView
    android:id="@+id/fragmentB"
    android:name="com.ptithcm.fragments.FragmentB"
    android:layout_width="match_parent"
    android:layout_weight="1"
    android:layout_height="wrap_content" />
```

```
View v;
```

```
v = inflater.inflate(R.layout.fragment_a, container, attachToRoot: false);
```

```
button = v.findViewById(R.id.button);
```

```
return v;
```

```
public class FragmentA extends Fragment {
```

```
    Button button; 2 usages
```

```
    int count = 0; 2 usages
```

```
    FragmentB.Counter counter; 2 usages
```

```
    button.setOnClickListener(new View.OnClickListener() {
```

```
        @Override
```

```
        public void onClick(View view) {
```

```
            count++;
```

```
            // Sử dụng Activity để giao tiếp fragment
```

```
            MainActivity mainActivity = (MainActivity) getActivity();
```

```
            mainActivity.incrementValue(count);
```

```
        }
```

```
    });
```

```

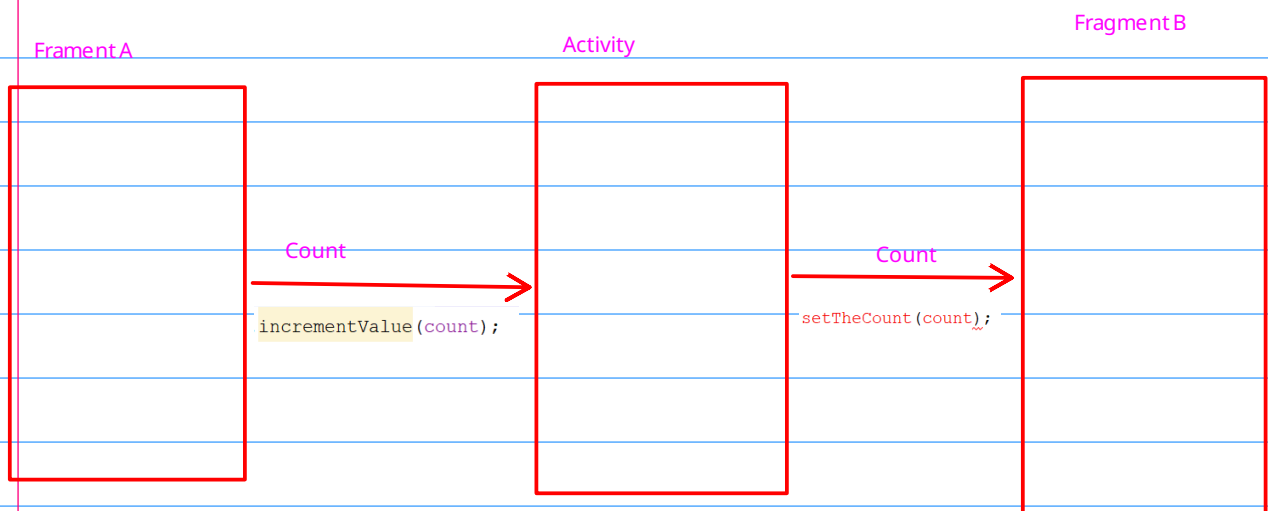
public void incrementValue(int count) { no usages 1 related problem
    FragmentB fragmentB;
    fragmentB = (FragmentB) getSupportFragmentManager().
    findFragmentById(R.id.fragmentB);
    if (fragmentB != null) {
        fragmentB.setTheCount(count);
    }
}

```

```

public void setTheCount(int count) { 1 usage
    textView.setText("Count: " + count);
}

```



```
public interface Counter{ 2 usages 1 implementation
    public void incrementValue(int count); 1 usage 1 implementation
}
```

```
FragmentB.Counter counter; 2 usages
```

```
// Sử dụng Activity để giao tiếp fragment
// MainActivity mainActivity = (MainActivity) getActivity();
// mainActivity.incrementValue(count);
// Sử dụng Interface
counter = (FragmentB.Counter) getActivity();
counter.incrementValue(count);
```

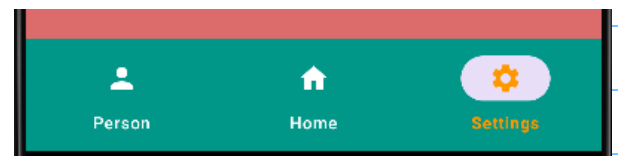
```
public class MainActivity extends AppCompatActivity implements FragmentB.Counter
{
    // public void incrementValue(int count) {
    //     FragmentB fragmentB;
    //     fragmentB = (FragmentB) getSupportFragmentManager().findFragmentById(R.id.fr
    //     if (fragmentB != null) {
    //         // fragmentB.setTheCount(count);
    //     }
    // }
    @Override 1 usage
    public void incrementValue(int count) {
        FragmentB fragmentB;
        fragmentB = (FragmentB) getSupportFragmentManager().
        findFragmentById(R.id.fragmentB);
        if (fragmentB != null) {
            fragmentB.setTheCount(count);
        }
    }
}
```

- © FirstFragment
- © MainActivity
- © SecondFragment
- © ThirdFragment

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:theme="@style/Base.Theme.ViewPagerNavigation"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```
<androidx.viewpager.widget.ViewPager
    android:id="@+id/view_pager"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_above="@id/bottom_navigation"/>
```

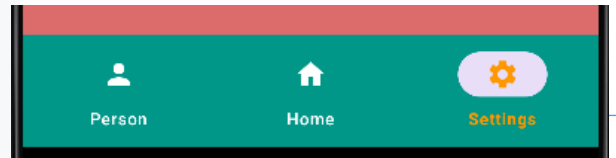
```
<com.google.android.material.bottomnavigation.BottomNavigationView
    android:id="@+id/bottom_navigation"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="#009688"
    android:layout_alignParentBottom="true"
    app:itemTextColor="@color/color_navigation"
    app:itemIconTint="@color/color_navigation"
    app:labelVisibilityMode="labeled"
    app:menu="@menu/menu_navigation"
/>
```



✓ res

✓ color

</> color_navigation.xml



```
<selector xmlns:android="http://schemas.android.com/apk/res/android">
    <item android:color="@color/white" android:state_checked="false" />
    <item android:color="@color/yellow" android:state_checked="true"/>
</selector>
```

✓ res

> color

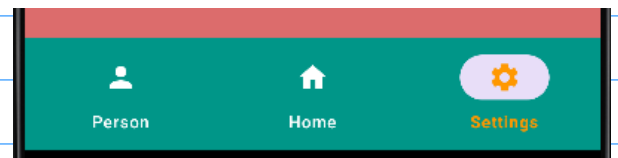
> drawable

> layout

✓ menu

</> menu_navigation.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/person"
        android:title="Person"
        android:icon="@drawable/person_white"/>
    <item
        android:id="@+id/home"
        android:title="Home"
        android:icon="@drawable/home_white"/>
    <item
        android:id="@+id/settings"
        android:title="Settings"
        android:icon="@drawable/settings_white"/>
</menu>
```



```

public class ViewpagerAdater extends FragmentStatePagerAdapter {
    @Override
    public Fragment getItem(int position) {
        switch (position) {
            case 0:
                return new FirstFragment();
            case 1:
                return new SecondFragment();
            case 2:
                return new ThirdFragment();
        }
        return null;
    }
    @Override
    > public int getCount() { return 3; }
}

```

ViewPager mViewPager; 7 usages

BottomNavigationView mBottomNavigationView; 5 usages

```

mViewPager = findViewById(R.id.view_pager);
mBottomNavigationView = findViewById(R.id.bottom_navigation);

```

```

ViewpagerAdater viewpagerAdater = new ViewpagerAdater(getSupportFragmentManager(),
    FragmentStatePagerAdapter.BEHAVIOR_RESUME_ONLY_CURRENT_FRAGMENT);
mViewPager.setAdapter(viewpagerAdater);
mViewPager.setCurrentItem(0);

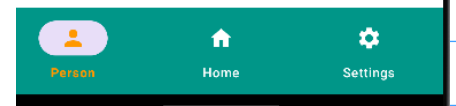
```

android.nonFinalResIds=false (fix! is switch case)

```

mViewPager.addOnPageChangeListener(new ViewPager.OnPageChangeListener() {
    @Override 3 usages
    public void onPageScrolled(int position, float positionOffset, int positionOffsetPixels) {
    }
    @Override 3 usages
    public void onPageSelected(int position) {
        switch (position) {
            case 0:
                mBottomNavigationView.getMenu().findItem(R.id.person).setChecked(true);
                //getSupportActionBar().setTitle("Person");
                break;
            case 1:
                mBottomNavigationView.getMenu().findItem(R.id.home).setChecked(true);
                //getSupportActionBar().setTitle("Home");
                break;
            case 2:
                mBottomNavigationView.getMenu().findItem(R.id.settings).setChecked(true);
                //getSupportActionBar().setTitle("Setting");
                break;
        }
    }
});

```



```

mBottomNavigationView.setOnNavigationItemSelectedListener(
    new BottomNavigationView.OnNavigationItemSelectedListener() {
        @Override 1 usage
        public boolean onNavigationItemSelected(@NonNull MenuItem item) {
            switch (item.getItemId()) {
                case R.id.person:
                    mViewPager.setCurrentItem(0);
                    //getSupportActionBar().setTitle("Person");
                    break;
                case R.id.home:
                    mViewPager.setCurrentItem(1);
                    //getSupportActionBar().setTitle("Home");
                    break;
                case R.id.settings:
                    mViewPager.setCurrentItem(2);
                    //getSupportActionBar().setTitle("Setting");
                    break;
            }
            return true;
        }
    });

```